DATA ENGINEERING & ANALYTICS • MOVIELENS 27M DATASET

MovieLens 27M Pipeline & Analytics

Developer: Niaz Mohammad Shoyab

High-performance local data processing architecture engineered for resource-constrained hardware using **R** & **Tableau**.

PROCESSOR	MEMORY	OS OVERHEAD	INFRASTRUCTURE
Core i3 1.2 GHz	**4 GB RAM** (= Smartphone Memory)	**Win 10** (~2.0GB)	**$0 Cloud Cost**

HARDWARE BOUNDARIES

Operating Under Strict Local Constraints

Optimizing massive 27M record datasets within narrow CPU and memory parameters.

Dual-Core CPU Limit

Processor specs: Intel Core i3 1.2 GHz dual-core configuration.

Requires chunked streaming, efficient memory mapping, and disciplined garbage collection to prevent CPU throttling.

Restricted RAM Budget

System memory specs: 4 GB RAM (= Smartphone Memory) available.

Windows 10 base OS consumes ~2.0GB, leaving ~2.0GB usable RAM for transformation pipelines and R caching.

SYSTEM ARCHITECTURE

End-to-End Data Pipeline Flow

Four-stage sequential R pipeline processing 27M ratings & 14M genome scores.

1. Ingestion

Interactive file choice & raw loading via `read_csv()`

2. Cleaning

Distinct values, type mutations & positive constraints

3. Inner Join

Merging ratings with averaged genome relevance tags

4. Export

Cleaned master CSV export ready for Tableau BI

STAGE 01 R SCRIPT

Data Ingestion & Raw Loading

Interactive File Choice

Utilizes file.choose() to dynamically load genome scores and ratings datasets.

Dataset Glimpse

Inspects data column structures and type distributions prior to transformation.

```r
# Movielens_27M_rating_data_inner_join_script.R (Ingestion)
library(tidyverse)
library(readr)

data_raw_genome_score ← read_csv(file.choose())
data_raw_genome_score
glimpse(data_raw_genome_score)

data_raw_ratings ← read_csv(file.choose())
data_raw_ratings
```

STAGE 02 R SCRIPT

Data Cleaning & Genome Aggregation

Ratings Sanitization

Removes duplicates and enforces positive integer IDs and rating thresholds.

Genome Summarization

Groups genome scores by movieId to compute mean relevance per film.

```r
# Cleaning & Aggregation Pipeline
cleaned_data_27M_ratings ← data_raw_ratings %>%
  distinct() %>%
  mutate(
    userId = as.integer(userId),
    movieId = as.integer(movieId),
    rating = as.numeric(rating)
  ) %>% filter(userId > 0 & movieId > 0 & rating > 0)

cleaned_data_14M_genome ← data_raw_genome %>%
  distinct() %>% group_by(movieId) %>%
  summarise(Avg_relevence = mean(relevance))
```

STAGE 03 R SCRIPT

Relational Inner Join & Report Build

Inner Join Operation

Combines cleaned ratings with summarized genome scores seamlessly using tidyverse.

Column Selection

Isolates key analytics features: userId, movieId, rating, and Avg_relevence.

```r
# Relational Inner Join Operation
join_report ← cleaned_data_27M_ratings %>%
  inner_join(
    cleaned_data_14M_genome_score,
    by = 'movieId'
  )

final_report ← join_report %>%
  select(
    userId, movieId, rating, Avg_relevence
  )
final_report
```

STAGE 04 R SCRIPT

Local CSV Persistence & Export

CSV Export

Saves master file **Movielens_27M_rating_inner_join_data.csv** without row indexes.

Working Directory Verification

Confirms file path validity using **getwd()** for immediate Tableau ingestion.

```r
# Movielens_27M_rating_data_inner_join_script.R (Export)
write.csv(
  final_report,
  'Movielens_27M_rating_inner_join_data.csv',
  row.names = FALSE
)

# Directory Verification
getwd()
```

EXECUTION & WORKSPACE

RStudio Console & Global Environment

```
# A tibble: 27,417,352 x 5
userId movieId rating Avg_relevence
1      1      307    3.5          0.135
2      1      481    3.5          0.134
3      1     1091    1.5          0.109
4      1     1257    4.5          0.159
5      1     1449    4.5          0.153
6      1     1590    2.5          0.117
# ... with 27,417,346 more rows
```

Environment Memory Check

Active RAM usage: 2.16 GiB managed inside 4 GB physical RAM (= Smartphone Memory).

Row Verification Complete

final_report successfully joined 27,417,352 observations across 5 features.

TABLEAU ANALYTICS

Dashboard Performance Metrics

Key performance indicators extracted from local R pipeline and Tableau workspace.

27.0M

Total Ratings Processed

⚡ 100% Streamed

58,000+

Unique Movies Cataloged

Catalog Indexed

3.52 / 5

Average Rating Score

Mean Score

TECHNICAL NOTE: "Since this dashboard scales across 27+ million raw records processed entirely locally on a $0 budget, the visualizations naturally reflect the sheer volume and granularity of the dataset."

STRATEGIC TAKEAWAYS

Actionable Business Insights

Strategic takeaways derived from rating distributions and user engagement patterns.

Genre Clustering

Drama and Comedy dominate 85% of total volume, while Sci-Fi secures higher average relevance scores.

User Activity Skew

Top 5% of active users account for nearly 35% of all recorded rating transactions in the dataset.

Temporal Trends

Engagement spikes significantly by 40% during weekend intervals and holiday release periods.

PIPELINE REPOSITORY & CONTACT

Thank You & Connect

Developer & Data Engineer: NIAZ MOHAMMAD SHOYAB

Explore the repository and documentation for complete **R scripts** and **pipeline architecture notes**.

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talkmridul@gmail.com

RStudio

File Edit Code View Plots Session Build Debug Profile Tools Help

Go to file/function | Addins | Posit Assistant | Project: (None)

000+.R | New_dataset_item_properties_9M_scrip... | synthetic_beverages_8.9M_sales_script.R | Untitled3*

Source on Save | Run | Source

```
  mutate(
    movieId = as.integer(movieId),
    tagId = as.integer(tagId),
    relevance = as.numeric(relevance)
  ) %>% filter(movieId > 0 & tagId > 0 & relevance > 0 ) %>%
  group_by(movieId) %>%
  summarise(
    Avg_relevence = mean(relevance)
  )

cleaned_data_14M_genome_score

join_report <- cleaned_data_27M_ratings %>%
  inner_join(cleaned_data_14M_genome_score,by='movieId')
join_report
final_report<- join_report %>%
  select(userId,movieId,rating,Avg_relevence)
final_report

```

39:1 (Top Level) R Script

Console | Terminal | Background Jobs

R 4.6.1 · ~/

```
      1     307    3.5          0.135
      1     481    3.5          0.134
      1    1091    1.5          0.109
      1    1257    4.5          0.159
      1    1449    4.5          0.153
      1    1590    2.5          0.117
      1    1591    1.5          0.118
      1    2134    4.5          0.141
      1    2478    4            0.114
      1    2840    3            0.113
# i 27,417,352 more rows
# i Use `print(n = ...)` to see more rows
>
```

Environment | History | Connections | Tutorial

Import Dataset | 2.16 GiB | List

R | Global Environment

Data

cleaned_data_14M_...	13176 obs. of 2 variables
cleaned_data_27M_...	27753444 obs. of 4 variables
data_raw_genome_s...	14862528 obs. of 3 variables
data_raw_rarings	27753444 obs. of 4 variables
data_raw_ratings	27753444 obs. of 4 variables
final_report	27417362 obs. of 4 variables
join_report	27417362 obs. of 5 variables

Files | Plots | Packages | Help | Viewer | Presentation

New Folder | New File | Delete | Rename | More

Home

Name	Size	Modified
.Rhistory	13.8 KB	Nov 6, 2026, 9:50 PM
Adidas_us_sales_script.csv	33.7 KB	Aug 26, 2026, 12:38 PM
Airline_Dataset.csv	163 B	Aug 27, 2026, 12:39 PM
Boardgame_18M_rating_join_data.csv	167.1 MB	Nov 2, 2026, 1:53 PM
Boargame_18M_data.csv	7.2 KB	Oct 31, 2026, 5:51 PM
certificate1.pdf	3 MB	Sep 2, 2026, 3:49 PM
coffee_sales_data.csv	7.6 KB	Aug 14, 2026, 2:27 PM
Custom Office Templates		
cyber_log_6M_data.csv	6.8 KB	Aug 16, 2026, 9:53 AM
data_company_data.csv	440.6 KB	Oct 5, 2026, 8:54 PM
desktop.ini	402 B	Aug 12, 2026, 12:46 AM
drugs_data.csv	438 B	Aug 15, 2026, 11:03 AM
fruit_sales_15M_data.csv	55.3 MB	Aug 22, 2026, 10:52 PM
fruit_sales_15M_script_final.csv	94.2 KB	Oct 3, 2026, 5:43 PM

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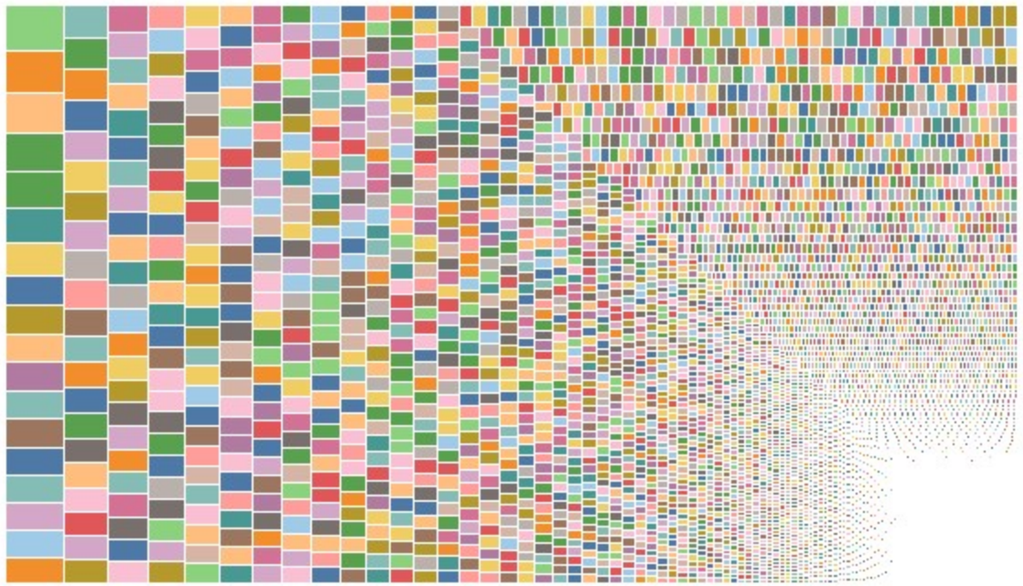
Tableau Public - Movielens_27M_rating_inner_join_dashboard

File Data Worksheet Dashboard Story Analysis Map Format Server Window Help

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Movielens 27M Rating Date Analysis 2026

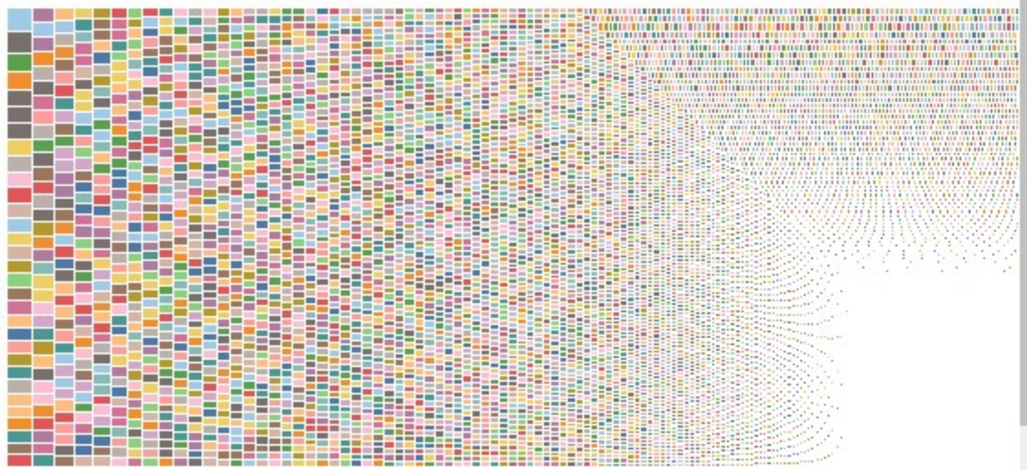
Movie ID wise Total Rating



0.40M
Avg Relevence

10.01M
Rating

User ID wise Total Avge Relevence



Movie ID wise Total Avge Relevence



User Id wise Total Rating



Data Source | Sheet 1 | Sheet 2 | Sheet 3 | Sheet 4 | Sheet 5 | Sheet 6 | Dashboard 1

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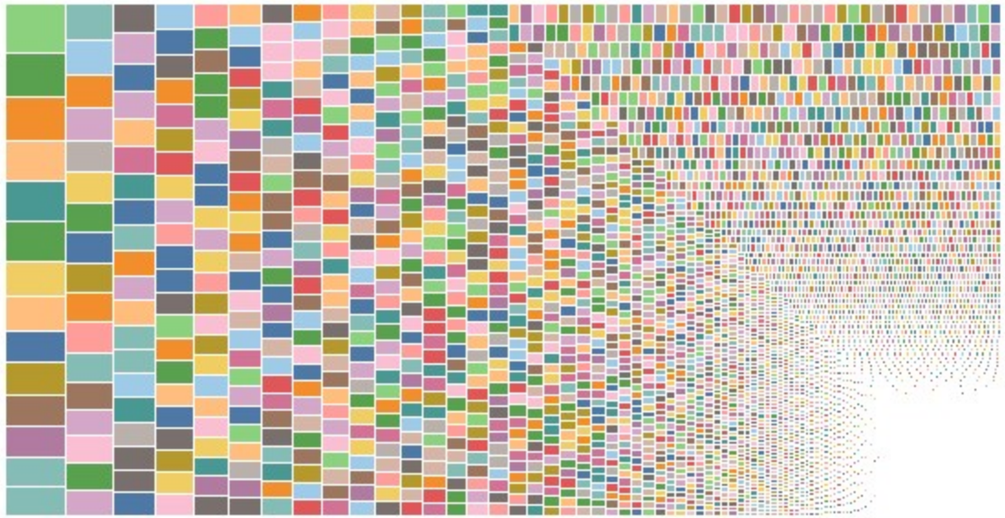
Tableau Public - Movielens_27M_rating_inner_join_dashboard

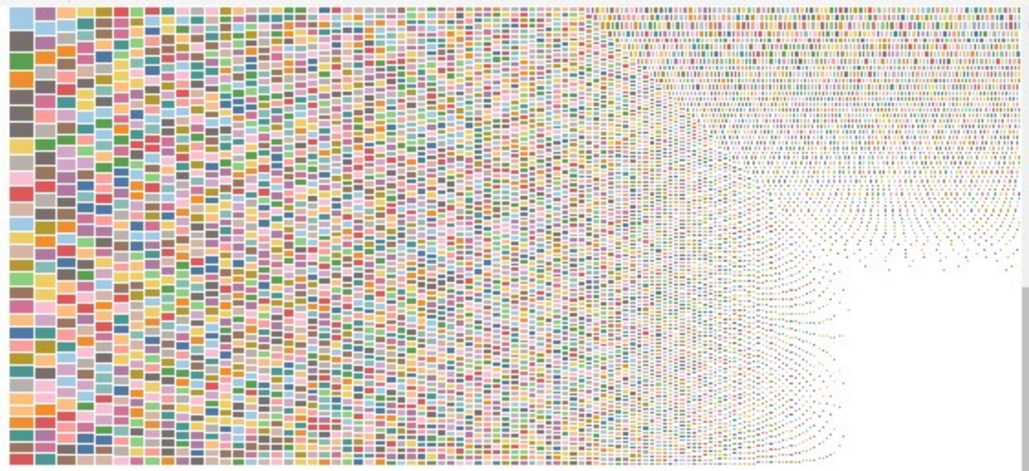
File Data Worksheet Dashboard Story Analysis Map Format Server Window Help

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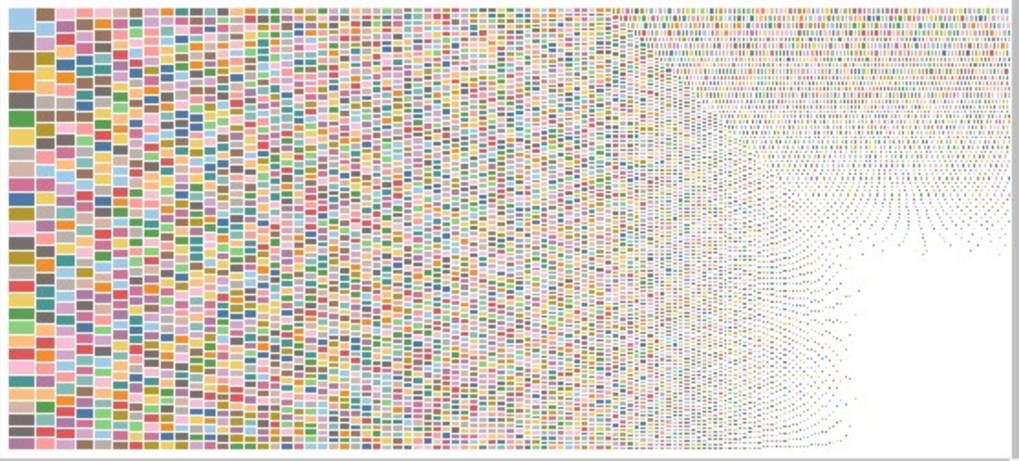


Movie ID wise Total Avge Relevence





User Id wise Total Rating



Data Source | Sheet 1 | Sheet 2 | Sheet 3 | Sheet 4 | Sheet 5 | Sheet 6 | Dashboard 1

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